

# git log/history

## References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

## Git History - Directory Commit Statistics

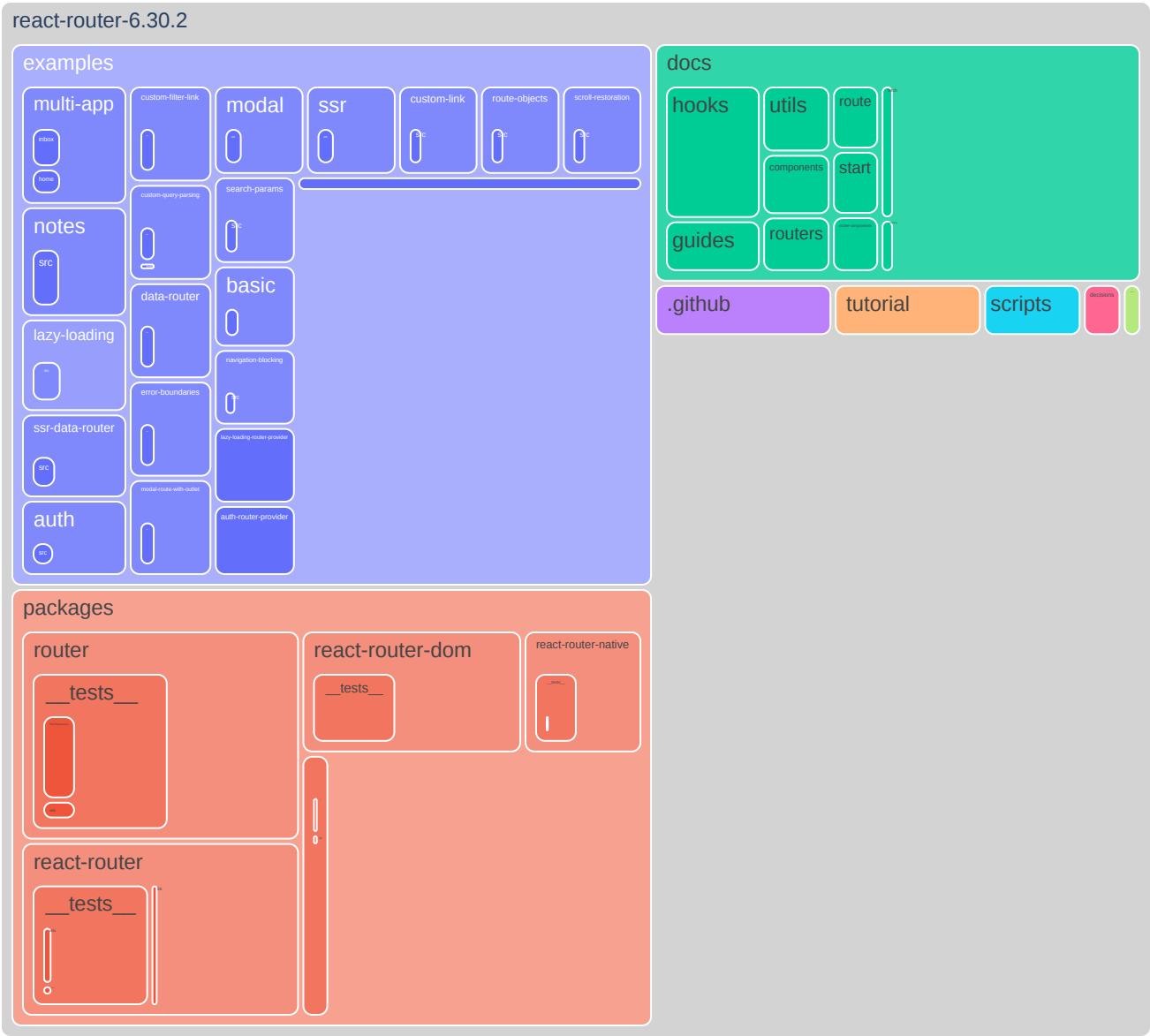
### Data Preview

|       | fileCount  | authorCount | commitCount | daysSinceLastCommit | daysSinceLastCreation | daysSinceLastModification |
|-------|------------|-------------|-------------|---------------------|-----------------------|---------------------------|
| count | 83.000000  | 83.000000   | 83.000000   | 83.000000           | 83.000000             | 83.000000                 |
| mean  | 25.253012  | 19.662651   | 130.060241  | 668.975904          | 1100.048193           | 669.975904                |
| std   | 77.782662  | 46.320010   | 293.488245  | 333.698892          | 401.928458            | 334.170176                |
| min   | 1.000000   | 2.000000    | 3.000000    | 25.000000           | 24.000000             | 24.000000                 |
| 25%   | 4.000000   | 4.000000    | 11.000000   | 432.000000          | 830.500000            | 431.000000                |
| 50%   | 10.000000  | 6.000000    | 26.000000   | 817.000000          | 1185.000000           | 816.000000                |
| 75%   | 13.000000  | 13.000000   | 75.000000   | 916.000000          | 1509.000000           | 915.000000                |
| max   | 633.000000 | 344.000000  | 1996.000000 | 1550.000000         | 1606.000000           | 1549.000000               |

|    | directoryPath                                                           | directoryParentPath                                        | directoryName  | fileCount | firstFile                                                                                          | mostF |
|----|-------------------------------------------------------------------------|------------------------------------------------------------|----------------|-----------|----------------------------------------------------------------------------------------------------|-------|
| 0  | react-router-6.30.2/examples/custom-query-parsing/types                 | react-router-6.30.2/examples/custom-query-parsing          | types          | 1         | react-router-6.30.2/examples/custom-query-parsing/types/jsurl.d.ts                                 |       |
| 1  | react-router-6.30.2/packages/react-router-dom-v5-compat/lib             | react-router-6.30.2/packages/react-router-dom-v5-compat    | lib            | 1         | react-router-6.30.2/packages/react-router-dom-v5-compat/lib/components.tsx                         |       |
| 2  | react-router-6.30.2/packages/react-router-dom/__tests__/polyfills       | react-router-6.30.2/packages/react-router-dom/__tests__    | polyfills      | 1         | react-router-6.30.2/packages/react-router-dom/__tests__/polyfills/drop-FormData-submitter.ts       |       |
| 3  | react-router-6.30.2/packages/react-router/__tests__/_snapshots__        | react-router-6.30.2/packages/react-router/__tests__        | __snapshots__  | 1         | react-router-6.30.2/packages/react-router/__tests__/_snapshots__/route-matching-test.tsx.snap      |       |
| 4  | react-router-6.30.2/patches                                             | react-router-6.30.2                                        | patches        | 1         | react-router-6.30.2/patches/@changesets_get-dependents-graph@1.3.6.patch                           |       |
| 5  | react-router-6.30.2/.changeset                                          | react-router-6.30.2                                        | .changeset     | 2         | react-router-6.30.2/.changeset/README.md                                                           |       |
| 6  | react-router-6.30.2/docs/upgrading                                      | react-router-6.30.2/docs                                   | upgrading      | 2         | react-router-6.30.2/docs/upgrading/future.md                                                       |       |
| 7  | react-router-6.30.2/examples/lazy-loading/src/pages                     | react-router-6.30.2/examples/lazy-loading/src              | pages          | 2         | react-router-6.30.2/examples/lazy-loading/src/pages/About.tsx                                      |       |
| 8  | react-router-6.30.2/.github/ISSUE_TEMPLATE                              | react-router-6.30.2/.github                                | ISSUE_TEMPLATE | 3         | react-router-6.30.2/.github/ISSUE_TEMPLATE/bug_report.yml                                          |       |
| 9  | react-router-6.30.2/examples/navigation-blocking/src                    | react-router-6.30.2/examples/navigation-blocking           | src            | 3         | react-router-6.30.2/examples/navigation-blocking/src/app.tsx                                       |       |
| 10 | react-router-6.30.2/packages/react-router-dom-v5-compat/__tests__       | react-router-6.30.2/packages/react-router-dom-v5-compat    | __tests__      | 3         | react-router-6.30.2/packages/react-router-dom-v5-compat/__tests__/compat-router-test.tsx           |       |
| 11 | react-router-6.30.2/decisions                                           | react-router-6.30.2                                        | decisions      | 4         | react-router-6.30.2/decisions/0001-use-blocker.md                                                  |       |
| 12 | react-router-6.30.2/examples/basic/src                                  | react-router-6.30.2/examples/basic                         | src            | 4         | react-router-6.30.2/examples/basic/src/App.tsx                                                     |       |
| 13 | react-router-6.30.2/examples/custom-link/src                            | react-router-6.30.2/examples/custom-link                   | src            | 4         | react-router-6.30.2/examples/custom-link/src/App.tsx                                               |       |
| 14 | react-router-6.30.2/examples/custom-query-parsing/src                   | react-router-6.30.2/examples/custom-query-parsing          | src            | 4         | react-router-6.30.2/examples/custom-query-parsing/src/App.tsx                                      |       |
| 15 | react-router-6.30.2/examples/multi-app/home                             | react-router-6.30.2/examples/multi-app                     | home           | 4         | react-router-6.30.2/examples/multi-app/home/App.jsx                                                |       |
| 16 | react-router-6.30.2/examples/route-objects/src                          | react-router-6.30.2/examples/route-objects                 | src            | 4         | react-router-6.30.2/examples/route-objects/src/App.tsx                                             |       |
| 17 | react-router-6.30.2/examples/scroll-restoration/src                     | react-router-6.30.2/examples/scroll-restoration            | src            | 4         | react-router-6.30.2/examples/scroll-restoration/src/app.tsx                                        |       |
| 18 | react-router-6.30.2/examples/search-params/src                          | react-router-6.30.2/examples/search-params                 | src            | 4         | react-router-6.30.2/examples/search-params/src/App.tsx                                             |       |
| 19 | react-router-6.30.2/packages/react-router-native/__tests__/_snapshots__ | react-router-6.30.2/packages/react-router-native/__tests__ | __snapshots__  | 4         | react-router-6.30.2/packages/react-router-native/__tests__/_snapshots__/deep-linking-test.tsx.snap |       |
| 20 | react-router-6.30.2/packages/react-router/lib                           | react-router-6.30.2/packages/react-router                  | lib            | 4         | react-router-6.30.2/packages/react-router/lib/components.tsx                                       |       |
| 21 | react-router-6.30.2/packages/router/__tests__/_utils__                  | react-router-6.30.2/packages/router/__tests__              | utils          | 4         | react-router-6.30.2/packages/router/__tests__/_utils__/custom-matchers.ts                          |       |
| 22 | react-router-6.30.2/tutorial/src                                        | react-router-6.30.2/tutorial                               | src            | 4         | react-router-6.30.2/tutorial/src/App.jsx                                                           |       |
| 23 | react-router-6.30.2/.github/workflows                                   | react-router-6.30.2/.github                                | workflows      | 5         | react-router-6.30.2/.github/workflows/deduplicate-lock-file.yml                                    |       |
| 24 | react-router-6.30.2/docs/fetch                                          | react-router-6.30.2/docs                                   | fetch          | 5         | react-router-6.30.2/docs/fetch/index.md                                                            |       |
| 25 | react-router-6.30.2/examples/auth/src                                   | react-router-6.30.2/examples/auth                          | src            | 5         | react-router-6.30.2/examples/auth/src/App.tsx                                                      |       |
| 26 | react-router-6.30.2/examples/custom-filter-link/src                     | react-router-6.30.2/examples/custom-filter-link            | src            | 5         | react-router-6.30.2/examples/custom-filter-link/src/App.tsx                                        |       |
| 27 | react-router-6.30.2/examples/data-router/src                            | react-router-6.30.2/examples/data-router                   | src            | 5         | react-router-6.30.2/examples/data-router/src/app.tsx                                               |       |
| 28 | react-router-6.30.2/examples/error-boundaries/src                       | react-router-6.30.2/examples/error-boundaries              | src            | 5         | react-router-6.30.2/examples/error-boundaries/src/app.tsx                                          |       |
| 29 | react-router-6.30.2/examples/modal-route-with-outlet/src                | react-router-6.30.2/examples/modal-route-with-outlet       | src            | 5         | react-router-6.30.2/examples/modal-route-with-outlet/src/App.tsx                                   |       |

# Number of files per directory

Directories and their file count



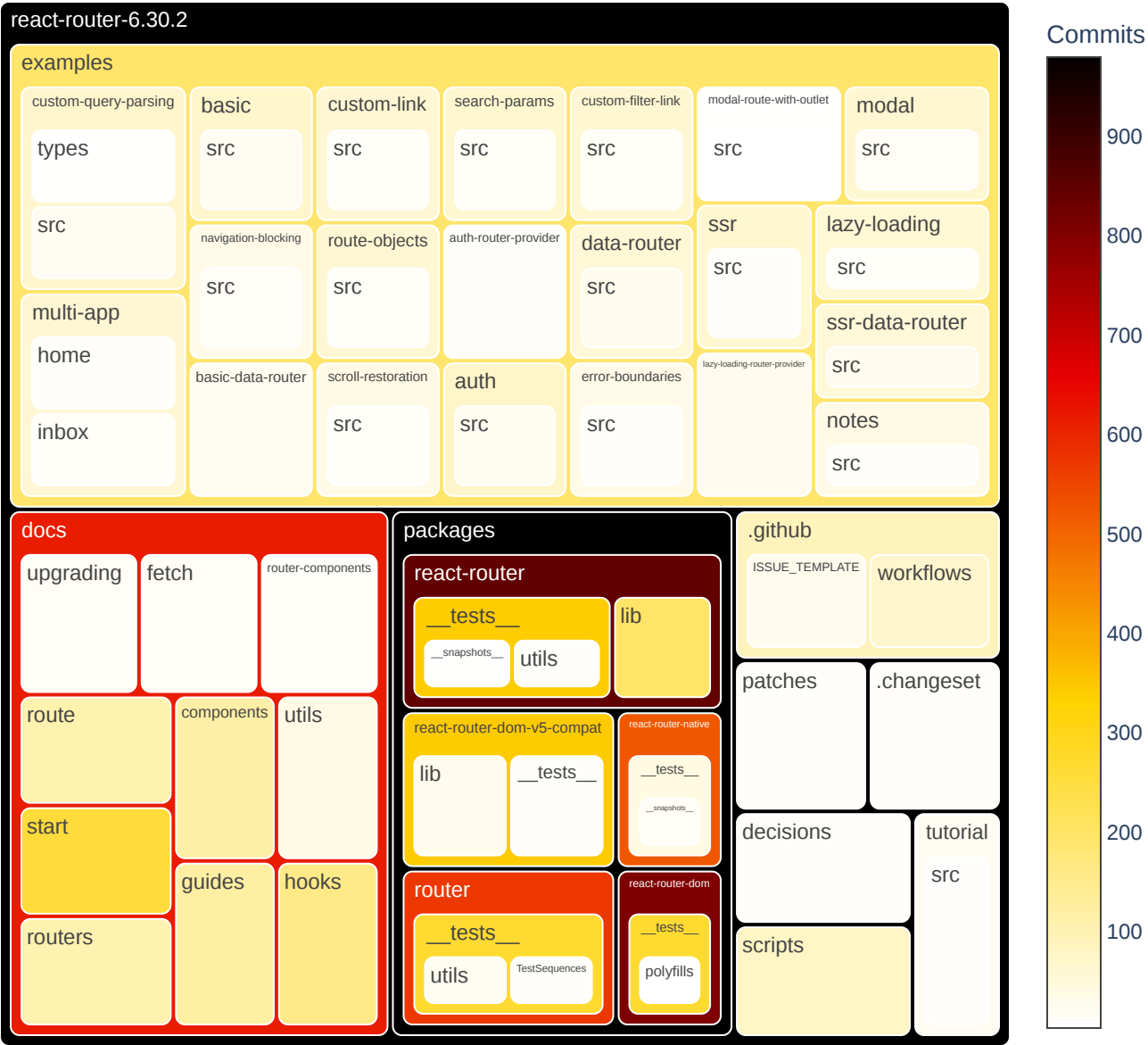
# Most frequent file extension per directory

Most frequent file extension per directory



Number of commits per directory

Number of git commits



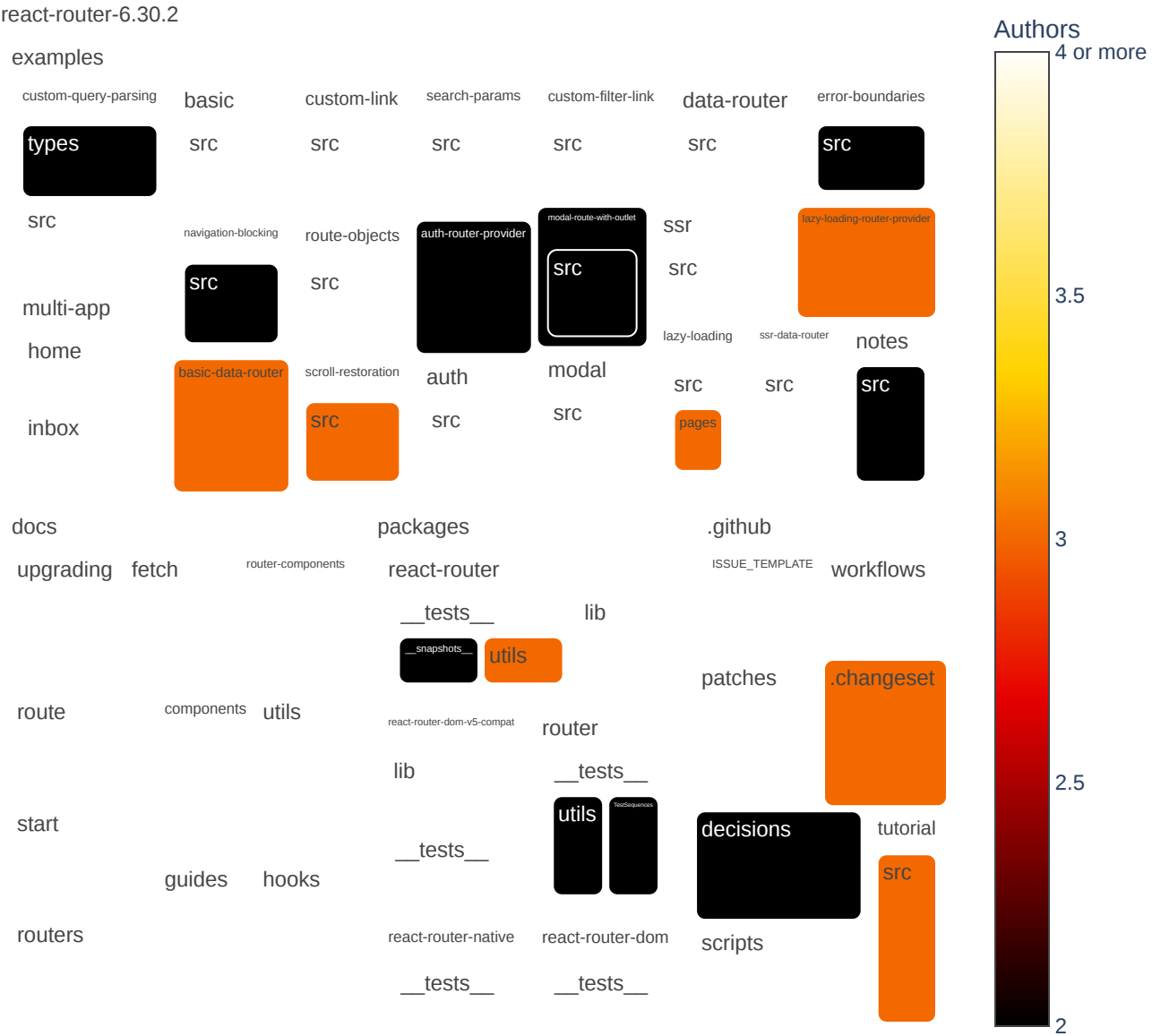
Number of distinct authors per directory

Number of distinct commit authors



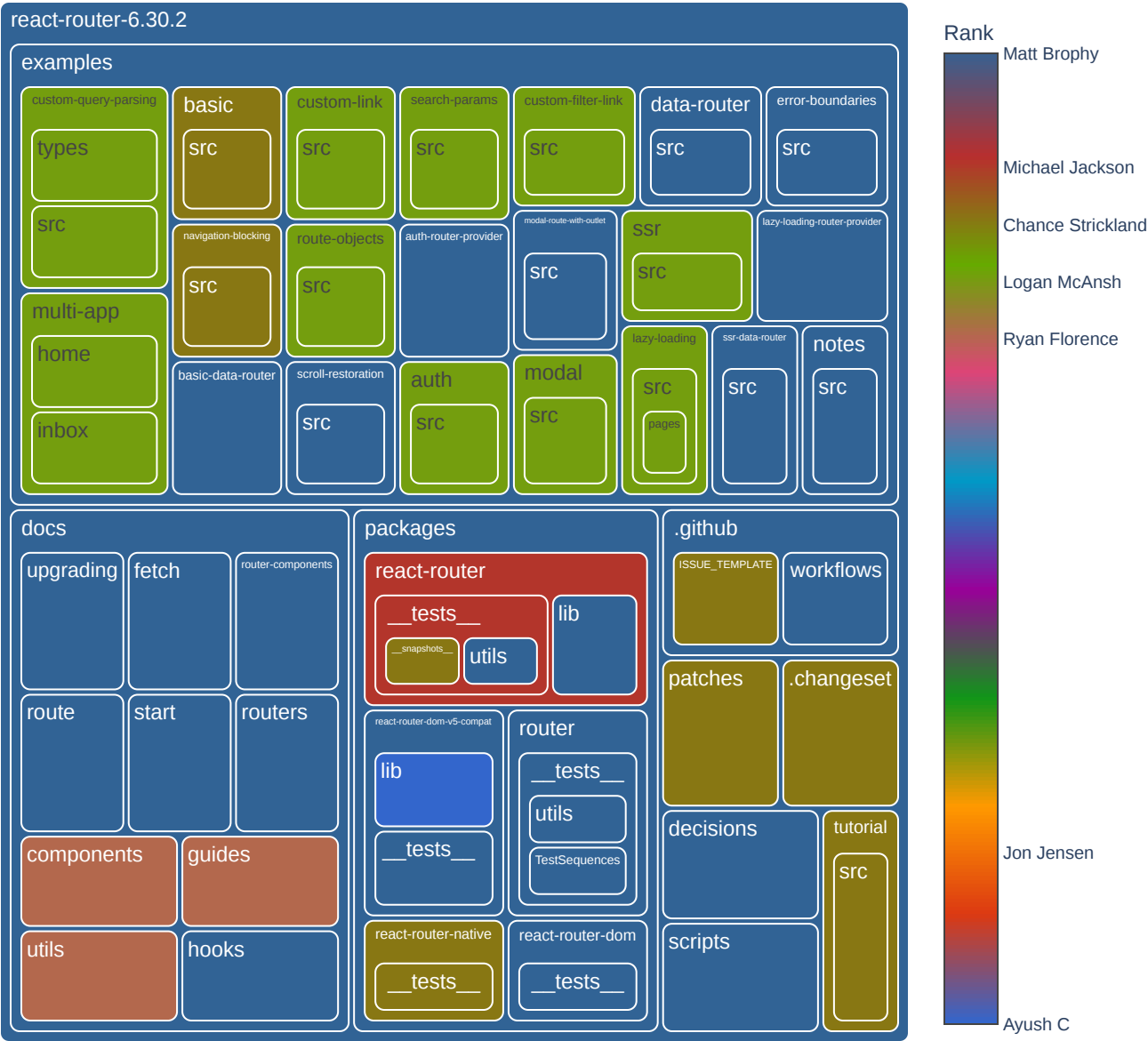
Directories with very few different authors

Number of distinct commit authors (red/black = only one or very few authors)



Main author per directory

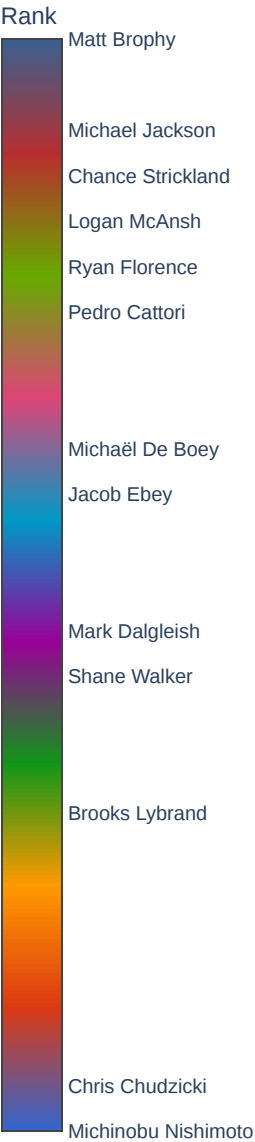
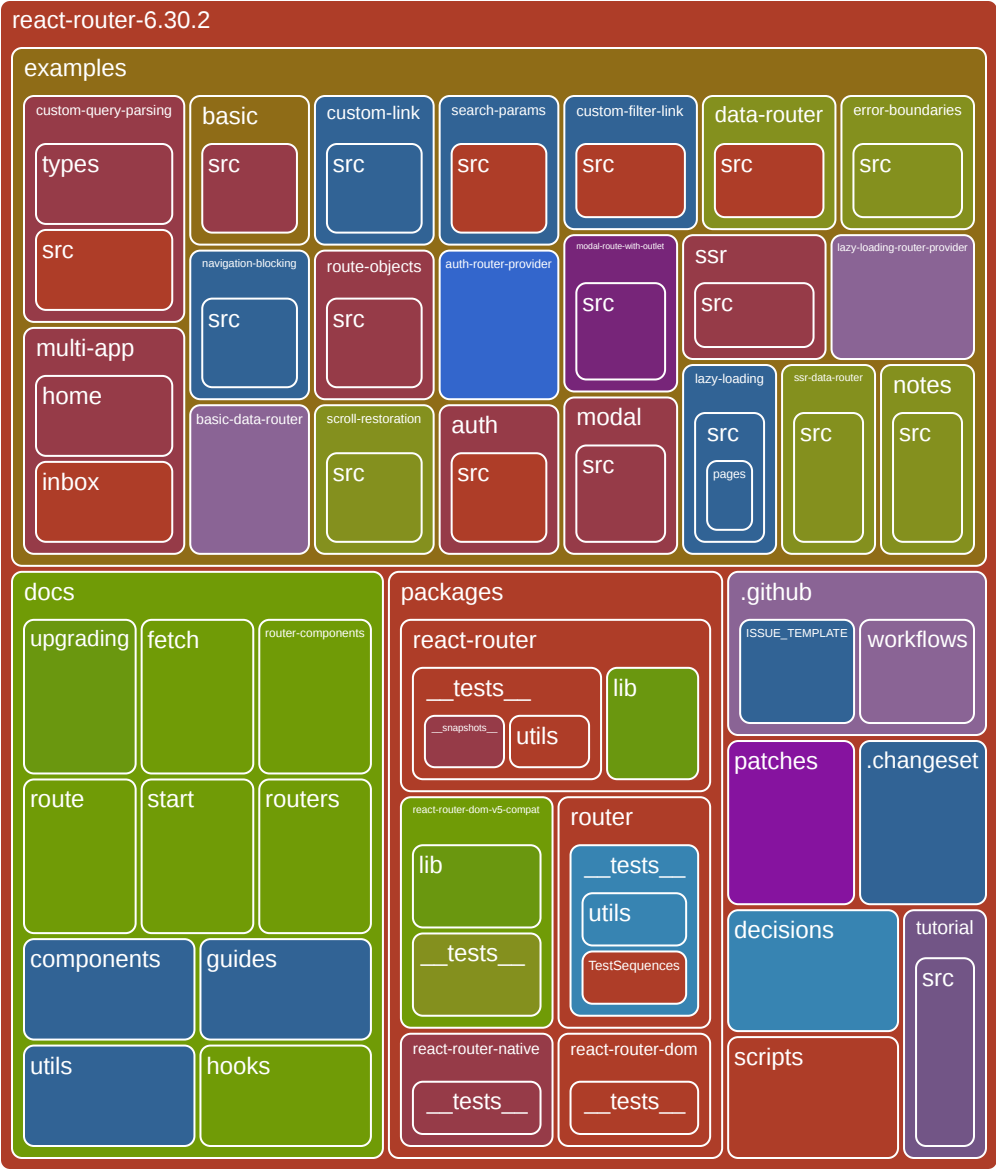
Main authors with highest number of commits



Second author per directory



Second author with the second highest number of commits



Days since last commit per directory

Days since last commit

react-router-6.30.2



Days since last commit per directory (ranked)

Rank of days since last commit

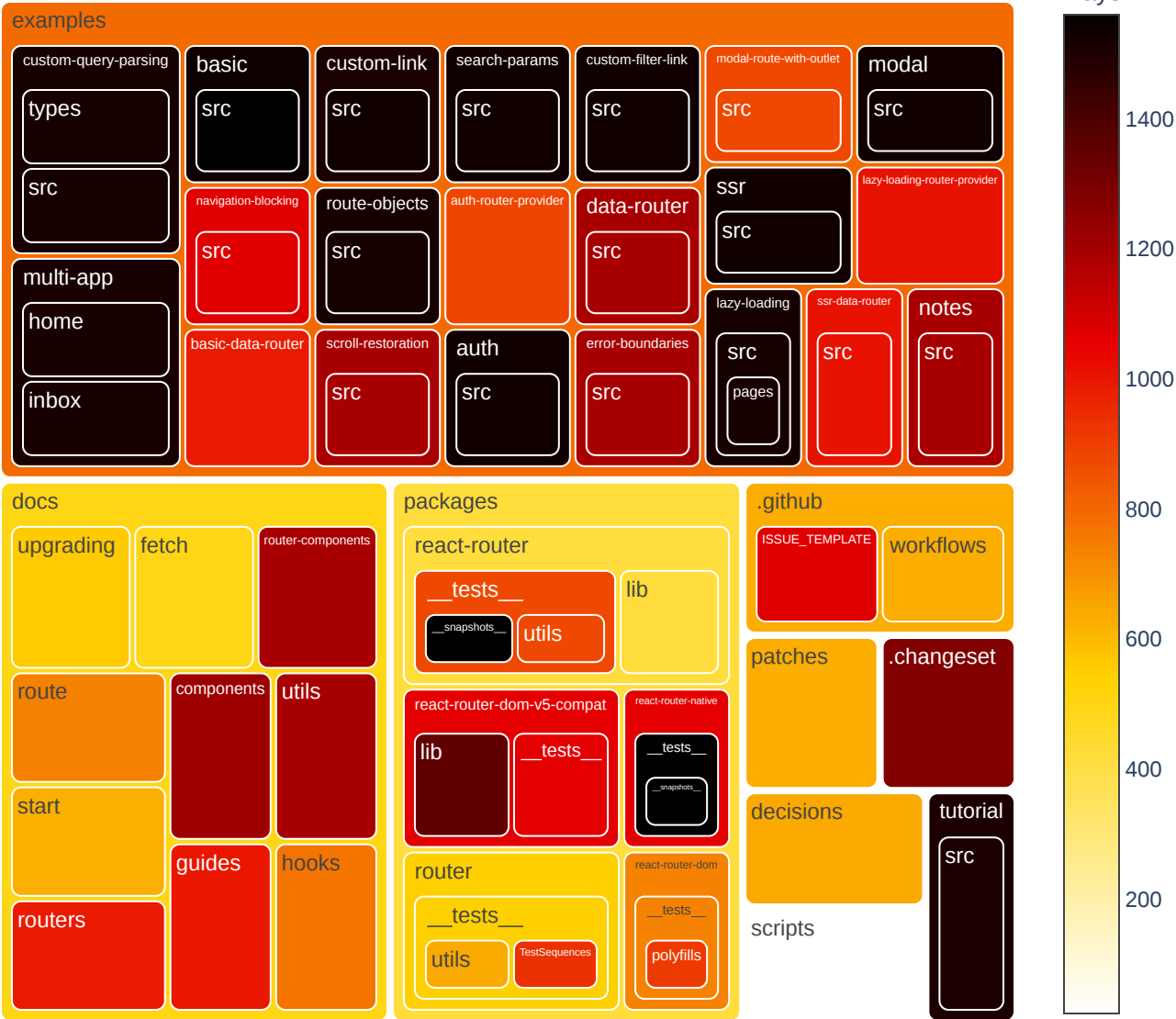
react-router-6.30.2



Days since last file creation per directory

Days since last file creation

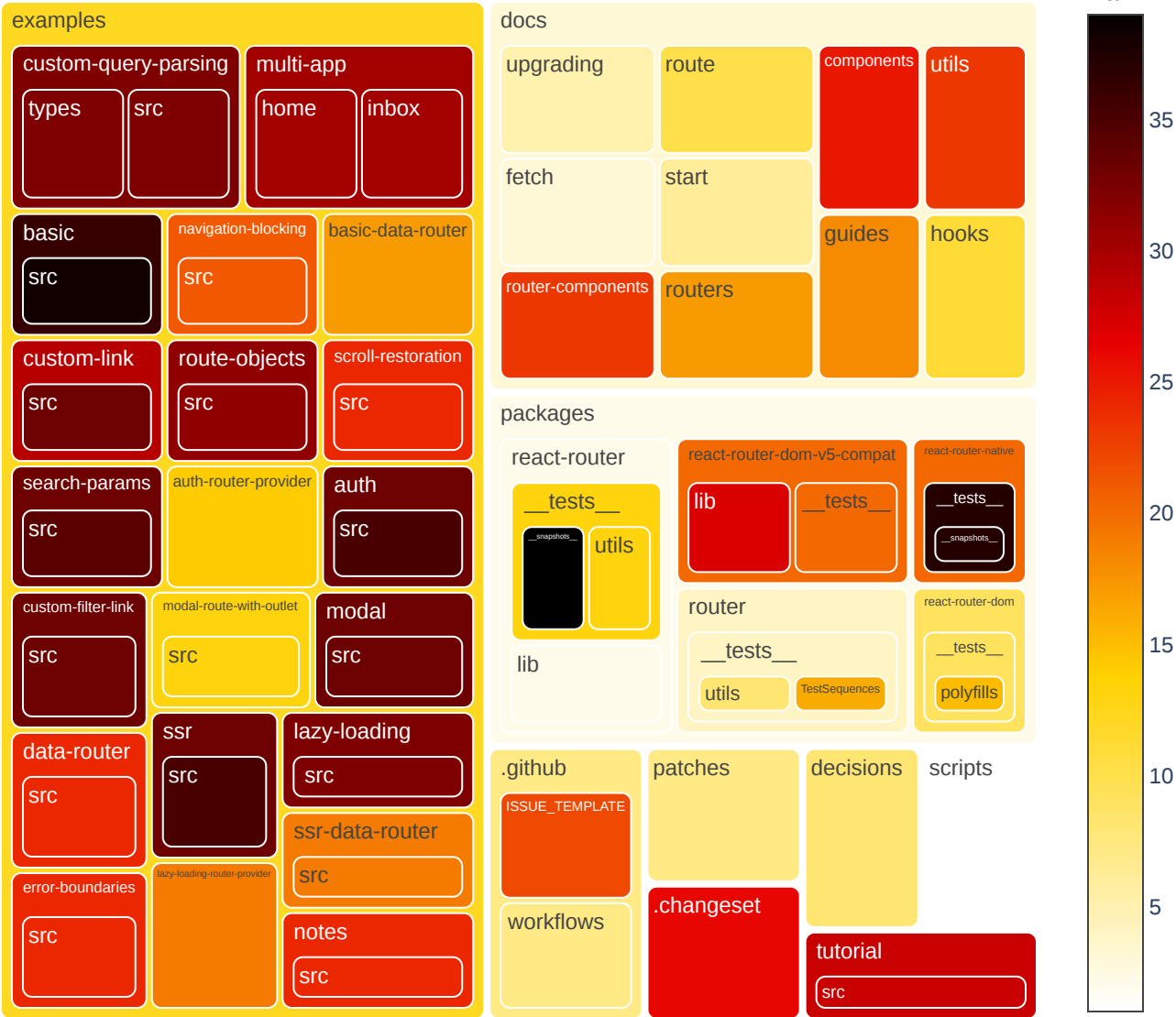
react-router-6.30.2



Days since last file creation per directory (ranked)

Rank of days since last file creation

react-router-6.30.2



Days since last file modification per directory

Days since last file modification

react-router-6.30.2



Days since last file modification per directory (ranked)

## Rank of days since last file modification

react-router-6.30.2



## Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

## Preview data

Sum of commits that changed more than 30 files (each) = 229

Max changed files with one commit = 498

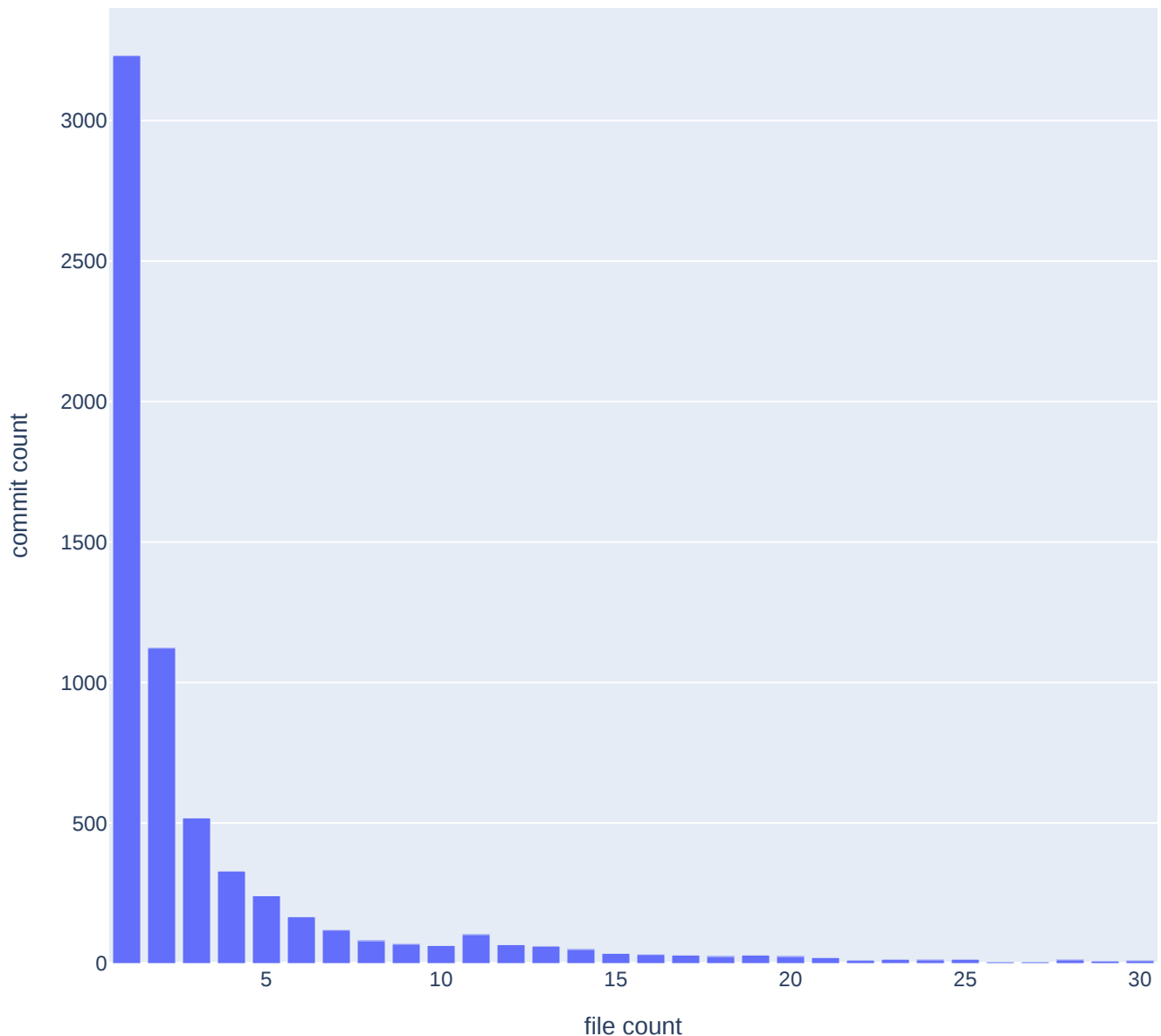
|       | filesPerCommit | commitCount |
|-------|----------------|-------------|
| count | 117.000000     | 117.000000  |
| mean  | 79.179487      | 57.692308   |
| std   | 78.464424      | 319.376992  |
| min   | 1.000000       | 1.000000    |
| 25%   | 30.000000      | 1.000000    |
| 50%   | 59.000000      | 3.000000    |
| 75%   | 96.000000      | 8.000000    |
| max   | 498.000000     | 3231.000000 |

|    | filesPerCommit | commitCount |
|----|----------------|-------------|
| 0  | 1              | 3231        |
| 1  | 2              | 1123        |
| 2  | 3              | 518         |
| 3  | 4              | 329         |
| 4  | 5              | 241         |
| 5  | 6              | 166         |
| 6  | 7              | 119         |
| 7  | 8              | 81          |
| 8  | 9              | 69          |
| 9  | 10             | 64          |
| 10 | 11             | 103         |
| 11 | 12             | 67          |
| 12 | 13             | 62          |
| 13 | 14             | 50          |
| 14 | 15             | 36          |
| 15 | 16             | 32          |
| 16 | 17             | 30          |
| 17 | 18             | 25          |
| 18 | 19             | 30          |
| 19 | 20             | 25          |
| 20 | 21             | 21          |
| 21 | 22             | 12          |
| 22 | 23             | 15          |
| 23 | 24             | 13          |
| 24 | 25             | 15          |
| 25 | 26             | 6           |
| 26 | 27             | 6           |
| 27 | 28             | 13          |
| 28 | 29             | 9           |
| 29 | 30             | 10          |

Bar chart with the number of files per commit distribution



## Changed files per commit



## Pairwise Changed Files

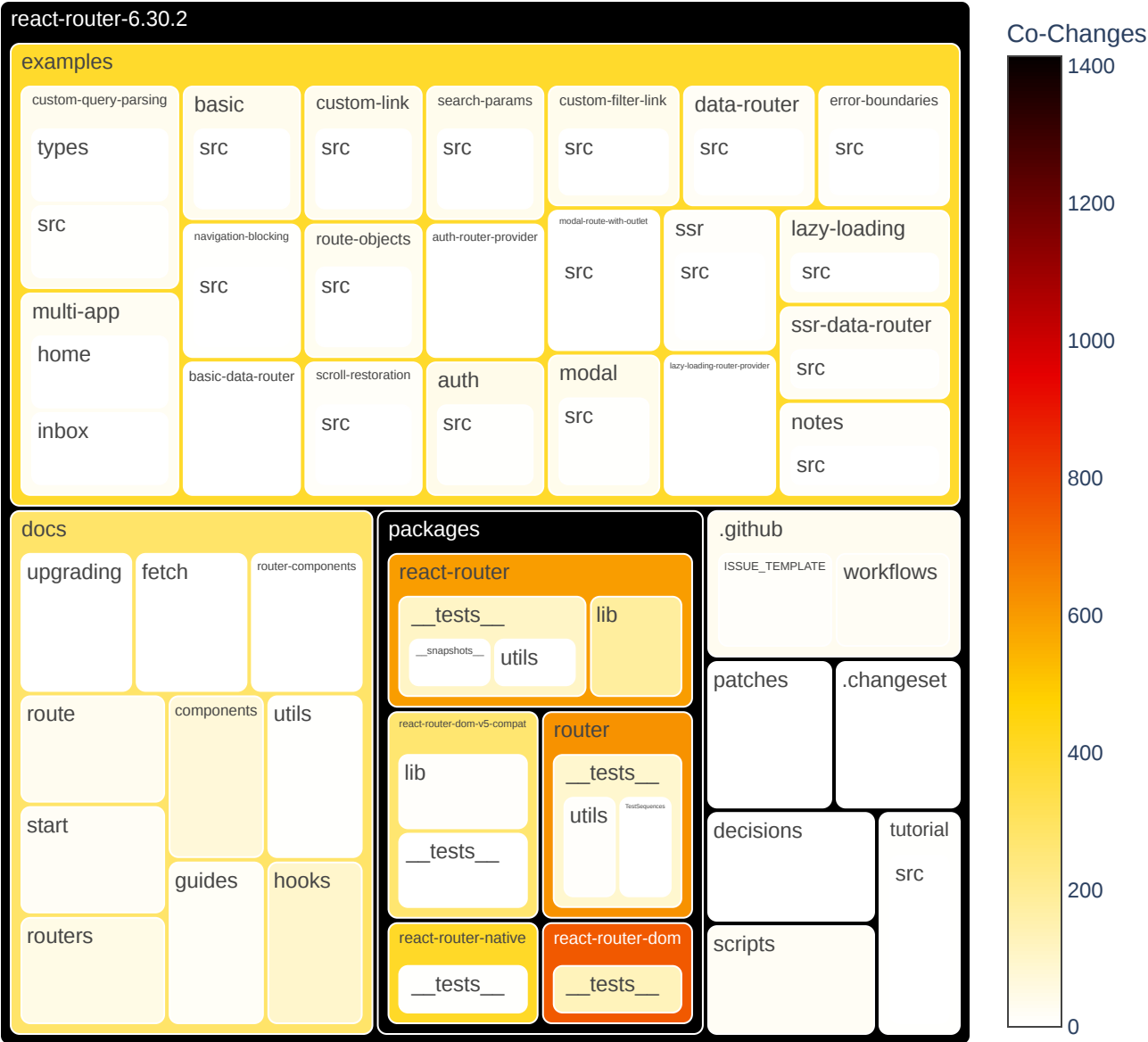
This section analyzes files that were changed together within the same commit and provides several metrics to quantify the strength of the co-change relationship:

- **Commit Count:** The number of commits in which two files were changed together.
- **Commit Lift:** A ratio that indicates whether the co-change pattern is stronger than random chance, given how often each file changes.
- **Jaccard Similarity:** The ratio of commits involving either file that also involved both files.

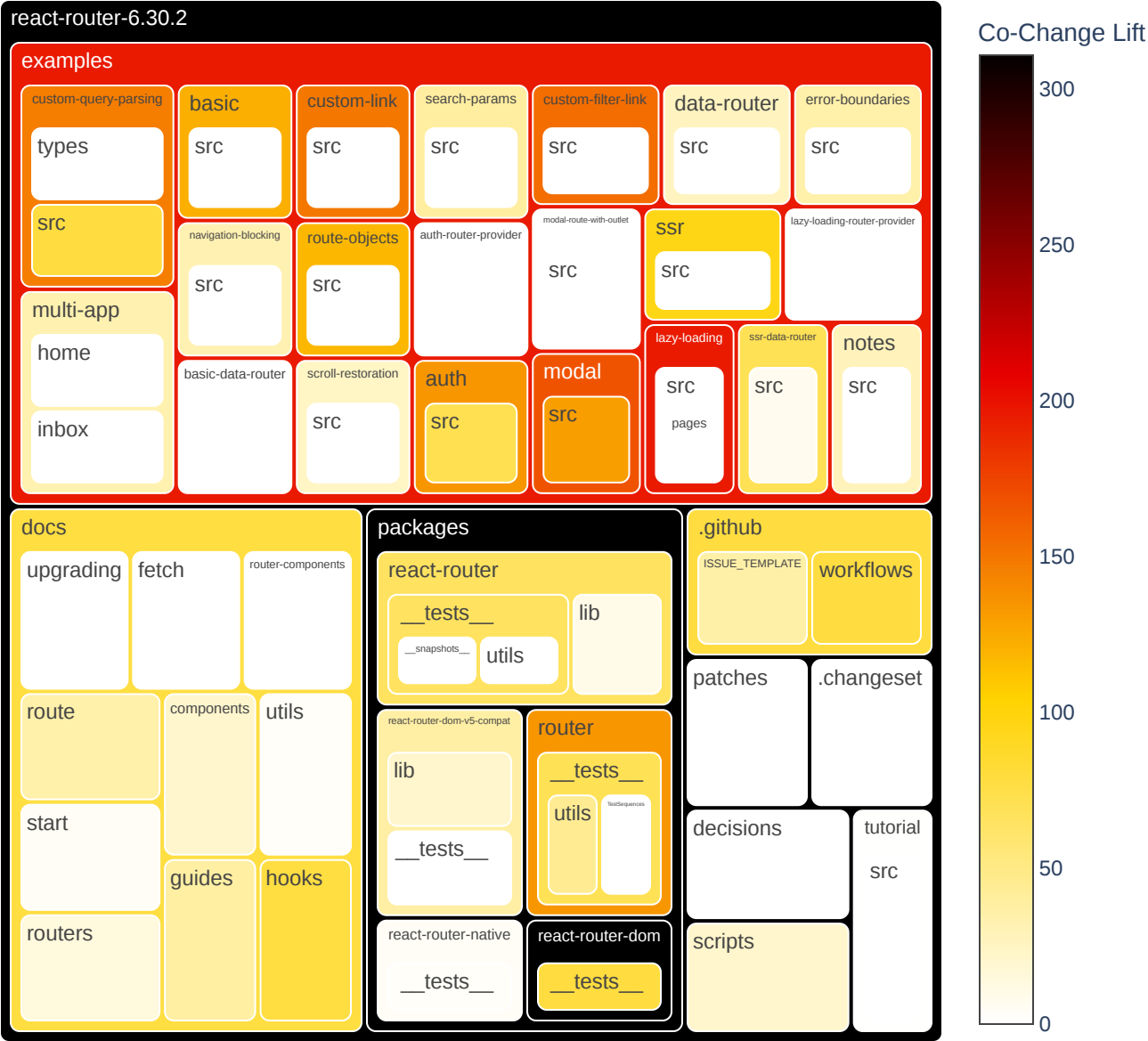
The following tables show the top pairwise changed files based on these metrics. The following charts show how these metrics are distributed across pairs of files that were changed together.

## Treemap with files changed frequently with others

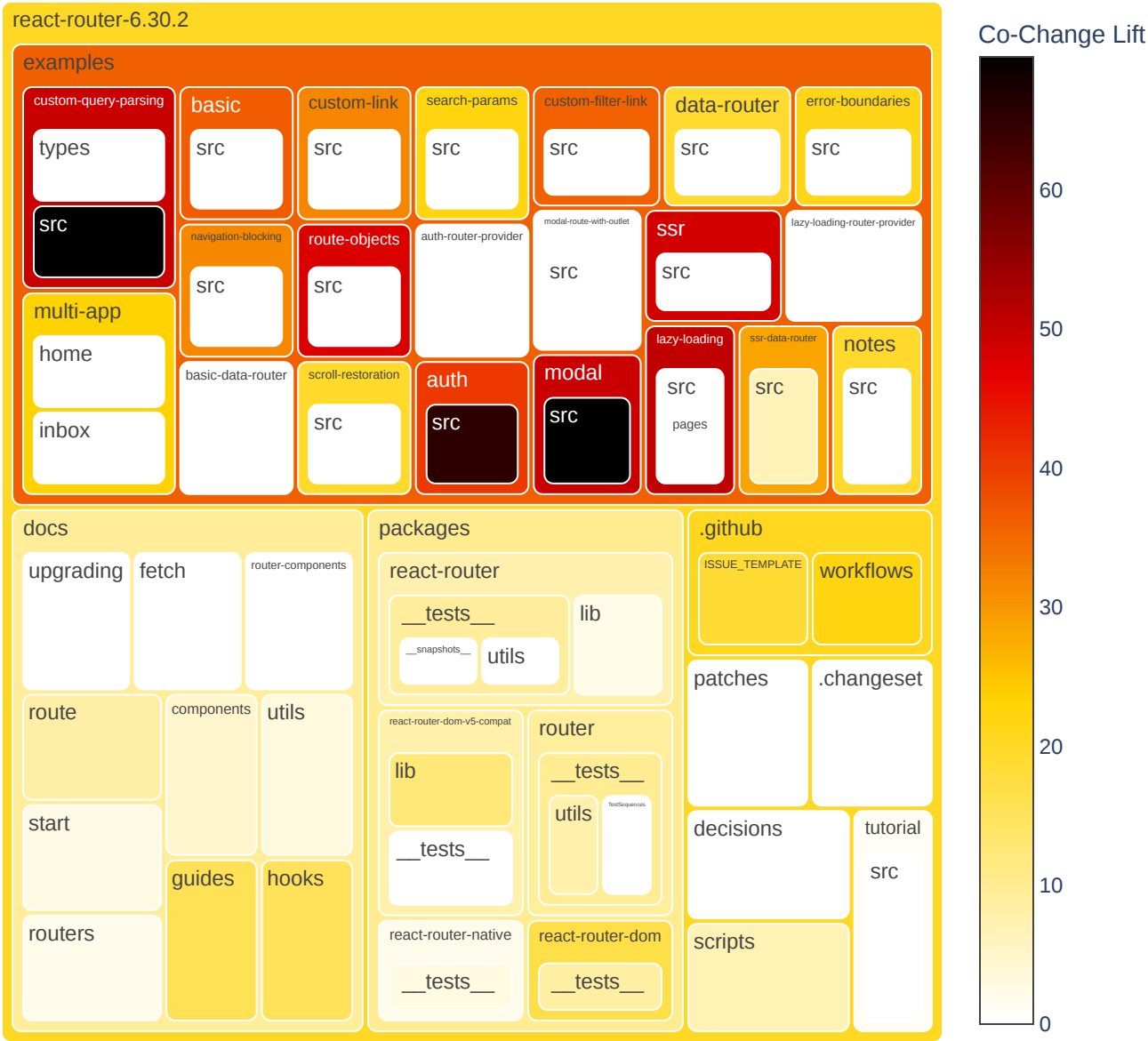
Files that likely co-change with others in update commits



Co-Changing files in update commits max lift (1=random, >1=more than random, <1=



Co-Changing files in update commits average lift (1=random, >1=more than random,

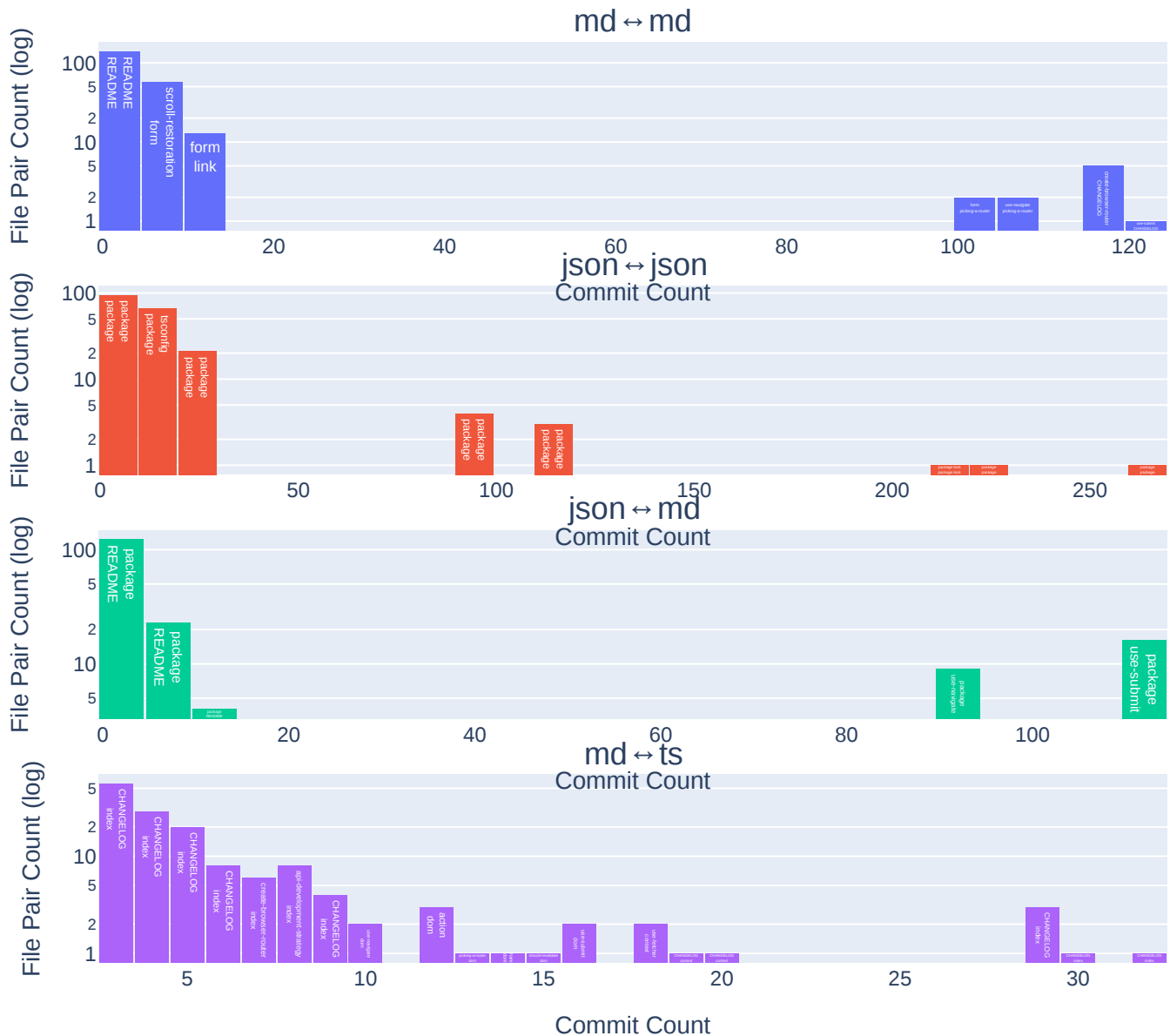


|   | fileExtensionPair | fileExtensionPairCount |
|---|-------------------|------------------------|
| 0 | md↔md             | 220                    |
| 1 | json↔json         | 192                    |
| 2 | json↔md           | 175                    |
| 3 | md↔ts             | 150                    |

Files changed together by commit count

|    | fileExtensionPair | updateCommitCount | GroupRank | filePair                      | filePairWithRelativePath                                                                          |
|----|-------------------|-------------------|-----------|-------------------------------|---------------------------------------------------------------------------------------------------|
| 0  | md↔md             | 123               | 1         | CHANGELOG↔CHANGELOG           | packages/react-router-dom/CHANGELOG.md↔packages/react-router/CHANGELOG.md                         |
| 1  | md↔md             | 117               | 2         | CHANGELOG↔CHANGELOG           | packages/react-router-dom-v5-compat/CHANGELOG.md↔packages/react-router-dom/CHANGELOG.md           |
| 2  | md↔md             | 108               | 3         | CHANGELOG↔CHANGELOG           | packages/react-router/CHANGELOG.md↔packages/router/CHANGELOG.md                                   |
| 3  | md↔md             | 106               | 4         | CHANGELOG↔CHANGELOG           | packages/react-router-dom/CHANGELOG.md↔packages/router/CHANGELOG.md                               |
| 4  | md↔md             | 100               | 5         | CHANGELOG↔CHANGELOG           | packages/react-router-native/CHANGELOG.md↔packages/router/CHANGELOG.md                            |
| 5  | md↔md             | 12                | 6         | use-fetcher↔use-submit        | docs/hooks/use-fetcher.md↔docs/hooks/use-submit.md                                                |
| 6  | md↔md             | 11                | 7         | link↔use-navigate             | docs/components/link.md↔docs/hooks/use-navigate.md                                                |
| 7  | md↔md             | 10                | 8         | form↔use-navigate             | docs/components/form.md↔docs/hooks/use-navigate.md                                                |
| 8  | md↔md             | 9                 | 9         | form↔use-submit               | docs/components/form.md↔docs/hooks/use-submit.md                                                  |
| 9  | md↔md             | 8                 | 10        | form↔link                     | docs/components/form.md↔docs/components/link.md                                                   |
| 10 | json↔json         | 264               | 1         | package↔package               | packages/react-router-dom/package.json↔packages/react-router/package.json                         |
| 11 | json↔json         | 220               | 2         | package↔package               | packages/react-router-dom/package.json↔packages/react-router-native/package.json                  |
| 12 | json↔json         | 215               | 3         | package↔package               | packages/react-router-native/package.json↔packages/react-router/package.json                      |
| 13 | json↔json         | 117               | 4         | package↔package               | packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom/package.json           |
| 14 | json↔json         | 98                | 5         | package↔package               | packages/react-router-dom-v5-compat/package.json↔packages/router/package.json                     |
| 15 | json↔json         | 97                | 6         | package↔package               | packages/react-router-dom/package.json↔packages/router/package.json                               |
| 16 | json↔json         | 21                | 7         | package↔package               | examples/auth/package.json↔examples/basic/package.json                                            |
| 17 | json↔json         | 20                | 8         | package↔package               | examples/basic/package.json↔examples/custom-filter-link/package.json                              |
| 18 | json↔json         | 19                | 9         | package↔package               | packages/react-router-native/package.json↔examples/auth/package.json                              |
| 19 | json↔json         | 18                | 10        | package↔package               | packages/react-router-dom/package.json↔examples/custom-filter-link/package.json                   |
| 20 | json↔md           | 112               | 1         | package↔CHANGELOG             | packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom/CHANGELOG.md           |
| 21 | json↔md           | 111               | 2         | package↔CHANGELOG             | packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.md |
| 22 | json↔md           | 94                | 3         | package↔CHANGELOG             | packages/router/package.json↔packages/react-router-dom/CHANGELOG.md                               |
| 23 | json↔md           | 93                | 4         | package↔CHANGELOG             | packages/router/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.md                     |
| 24 | json↔md           | 11                | 5         | package↔DEVELOPMENT           | packages/react-router/package.json↔DEVELOPMENT.md                                                 |
| 25 | json↔md           | 9                 | 6         | package↔create-browser-router | packages/react-router-dom-v5-compat/package.json↔docs/routers/create-browser-router.md            |
| 26 | json↔md           | 8                 | 7         | package↔tutorial              | packages/react-router-native/package.json↔docs/start/tutorial.md                                  |
| 27 | json↔md           | 6                 | 8         | package↔use-navigate          | packages/react-router-dom/package.json↔docs/hooks/use-navigate.md                                 |
| 28 | json↔md           | 5                 | 9         | package↔link                  | packages/router/package.json↔docs/components/link.md                                              |
| 29 | json↔md           | 4                 | 10        | package↔form                  | packages/react-router/package.json↔docs/components/form.md                                        |
| 30 | md↔ts             | 32                | 1         | CHANGELOG↔router              | packages/react-router/CHANGELOG.md↔packages/router/router.ts                                      |
| 31 | md↔ts             | 30                | 2         | CHANGELOG↔router              | packages/router/CHANGELOG.md↔packages/router/router.ts                                            |
| 32 | md↔ts             | 29                | 3         | CHANGELOG↔router              | packages/react-router-dom/CHANGELOG.md↔packages/router/router.ts                                  |
| 33 | md↔ts             | 20                | 4         | CHANGELOG↔utils               | packages/react-router/CHANGELOG.md↔packages/router/utils.ts                                       |
| 34 | md↔ts             | 19                | 5         | CHANGELOG↔utils               | packages/router/CHANGELOG.md↔packages/router/utils.ts                                             |
| 35 | md↔ts             | 18                | 6         | CHANGELOG↔utils               | packages/react-router-dom/CHANGELOG.md↔packages/router/utils.ts                                   |
| 36 | md↔ts             | 16                | 7         | CHANGELOG↔utils               | packages/react-router-dom-v5-compat/CHANGELOG.md↔packages/router/utils.ts                         |
| 37 | md↔ts             | 15                | 8         | form↔router                   | docs/components/form.md↔packages/router/router.ts                                                 |
| 38 | md↔ts             | 14                | 9         | use-submit↔router             | docs/hooks/use-submit.md↔packages/router/router.ts                                                |
| 39 | md↔ts             | 13                | 10        | use-submit↔dom                | docs/hooks/use-submit.md↔packages/react-router-dom/dom.ts                                         |

## Co-Changed Files by their commit count

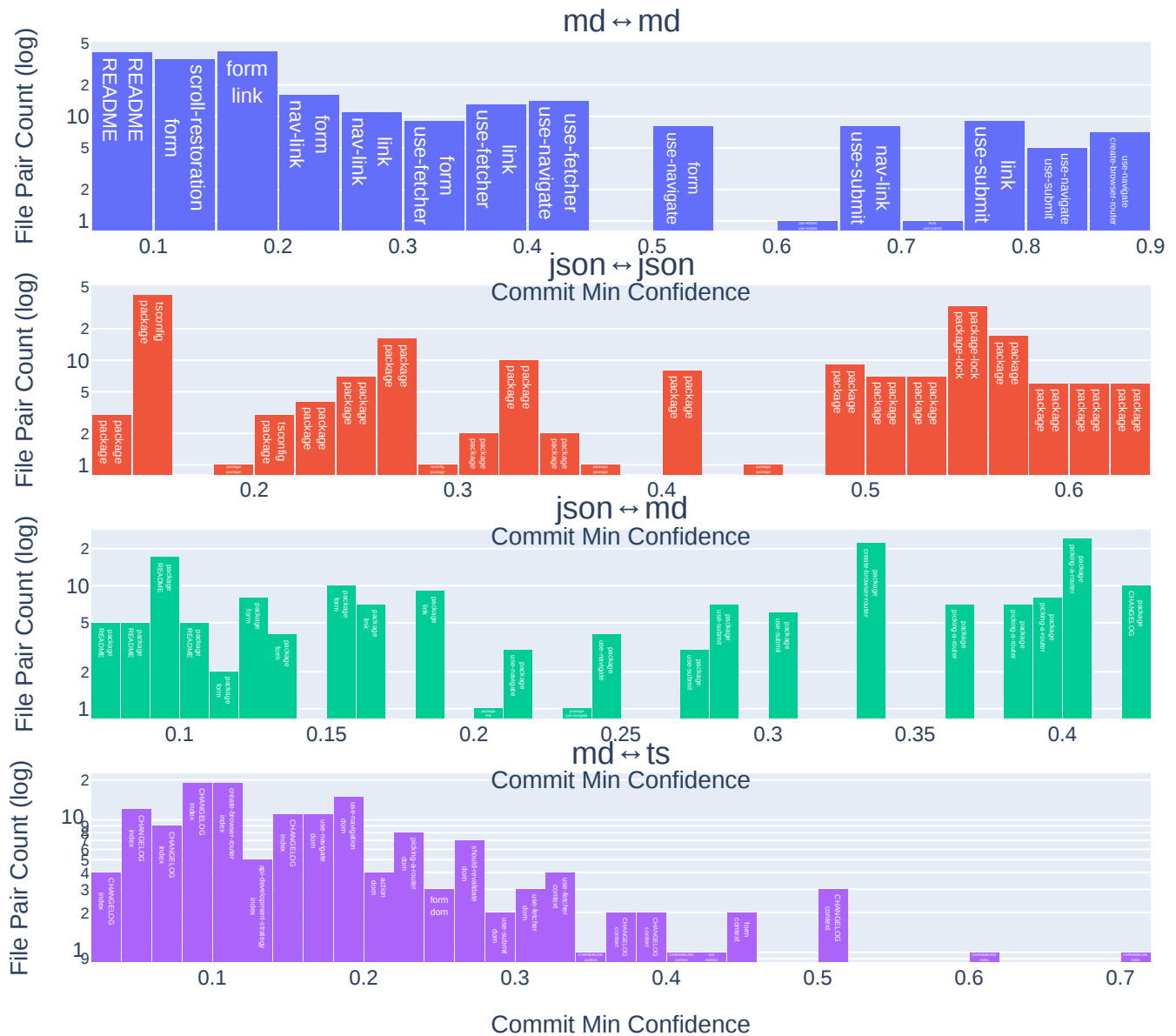


## Files changed together by commit min confidence

The commit min confidence is the commit count where both files were changed divided by the commit count of the file with the least commits. This metric is useful to identify pairs of files that are frequently changed together and is not biased by single files that are changed very often.

|    | fileExtensionPair | updateCommitMinConfidence | GroupRank | filePair                                 | filePairWithRelativePath                                                                          |
|----|-------------------|---------------------------|-----------|------------------------------------------|---------------------------------------------------------------------------------------------------|
| 0  | md↔md             | 0.888889                  | 1         | README↔README                            | examples/custom-filter-link/README.md↔examples/custom-link/README.md                              |
| 1  | md↔md             | 0.857143                  | 2         | README↔README                            | examples/custom-filter-link/README.md↔examples/modal/README.md                                    |
| 2  | md↔md             | 0.800000                  | 3         | README↔README                            | examples/ssr/README.md↔examples/auth/README.md                                                    |
| 3  | md↔md             | 0.750000                  | 4         | README↔README                            | examples/auth/README.md↔examples/route-objects/README.md                                          |
| 4  | md↔md             | 0.727273                  | 5         | README↔README                            | examples/ssr/README.md↔examples/basic/README.md                                                   |
| 5  | md↔md             | 0.666667                  | 6         | README↔README                            | examples/auth/README.md↔examples/custom-filter-link/README.md                                     |
| 6  | md↔md             | 0.600000                  | 7         | link↔use-view-transition-state           | docs/components/link.md↔docs/hooks/use-view-transition-state.md                                   |
| 7  | md↔md             | 0.500000                  | 8         | README↔README                            | examples/custom-query-parsing/README.md↔examples/lazy-loading/README.md                           |
| 8  | md↔md             | 0.446281                  | 9         | CHANGELOG↔CHANGELOG                      | packages/react-router/CHANGELOG.md↔packages/router/CHANGELOG.md                                   |
| 9  | md↔md             | 0.444444                  | 10        | use-route-loader-data↔picking-a-router   | docs/hooks/use-route-loader-data.md↔docs/routers/picking-a-router.md                              |
| 10 | json↔json         | 0.625000                  | 1         | package↔package                          | examples/basic/package.json↔examples/custom-filter-link/package.json                              |
| 11 | json↔json         | 0.600000                  | 2         | package↔package                          | examples/auth/package.json↔examples/basic/package.json                                            |
| 12 | json↔json         | 0.588235                  | 3         | package↔package                          | examples/auth/package.json↔examples/custom-link/package.json                                      |
| 13 | json↔json         | 0.581498                  | 4         | package↔package                          | packages/react-router-dom/package.json↔packages/react-router/package.json                         |
| 14 | json↔json         | 0.571429                  | 5         | package↔package                          | examples/auth/package.json↔examples/modal/package.json                                            |
| 15 | json↔json         | 0.566667                  | 6         | package↔package                          | examples/auth/package.json↔examples/multi-app/package.json                                        |
| 16 | json↔json         | 0.562500                  | 7         | package↔package                          | packages/react-router-dom/package.json↔examples/custom-filter-link/package.json                   |
| 17 | json↔json         | 0.548387                  | 8         | package↔package                          | examples/custom-filter-link/package.json↔examples/custom-query-parsing/package.json               |
| 18 | json↔json         | 0.542857                  | 9         | package↔package                          | packages/react-router-native/package.json↔examples/auth/package.json                              |
| 19 | json↔json         | 0.529412                  | 10        | package↔package                          | packages/react-router-native/package.json↔examples/custom-link/package.json                       |
| 20 | json↔md           | 0.428571                  | 1         | package↔README                           | examples/basic/package.json↔examples/modal/README.md                                              |
| 21 | json↔md           | 0.400722                  | 2         | package↔CHANGELOG                        | packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.md |
| 22 | json↔md           | 0.400000                  | 3         | package↔CHANGELOG                        | packages/router/package.json↔packages/react-router-dom/CHANGELOG.md                               |
| 23 | json↔md           | 0.395745                  | 4         | package↔CHANGELOG                        | packages/router/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.md                     |
| 24 | json↔md           | 0.392982                  | 5         | package↔CHANGELOG                        | packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom/CHANGELOG.md           |
| 25 | json↔md           | 0.391608                  | 6         | package↔CHANGELOG                        | packages/react-router-dom-v5-compat/package.json↔packages/react-router/CHANGELOG.md               |
| 26 | json↔md           | 0.388430                  | 7         | package↔CHANGELOG                        | packages/react-router-dom/package.json↔packages/router/CHANGELOG.md                               |
| 27 | json↔md           | 0.386207                  | 8         | package↔CHANGELOG                        | packages/react-router-native/package.json↔packages/react-router/CHANGELOG.md                      |
| 28 | json↔md           | 0.363636                  | 9         | package↔README                           | packages/react-router/package.json↔examples/basic/README.md                                       |
| 29 | json↔md           | 0.333333                  | 10        | package↔use-route-loader-data            | packages/router/package.json↔docs/hooks/use-route-loader-data.md                                  |
| 30 | md↔ts             | 0.714286                  | 1         | create-static-handler↔router             | docs/routers/create-static-handler.md↔packages/router/router.ts                                   |
| 31 | md↔ts             | 0.600000                  | 2         | should-revalidate↔should-revalidate-test | docs/route/should-revalidate.md↔packages/router/__tests__/should-revalidate-test.ts               |
| 32 | md↔ts             | 0.500000                  | 3         | README↔router                            | examples/ssr-data-router/README.md↔packages/router/router.ts                                      |
| 33 | md↔ts             | 0.444444                  | 4         | use-route-loader-data↔utils              | docs/hooks/use-route-loader-data.md↔packages/router/utils.ts                                      |
| 34 | md↔ts             | 0.428571                  | 5         | create-static-handler↔ssr-test           | docs/routers/create-static-handler.md↔packages/router/__tests__/ssr-test.ts                       |

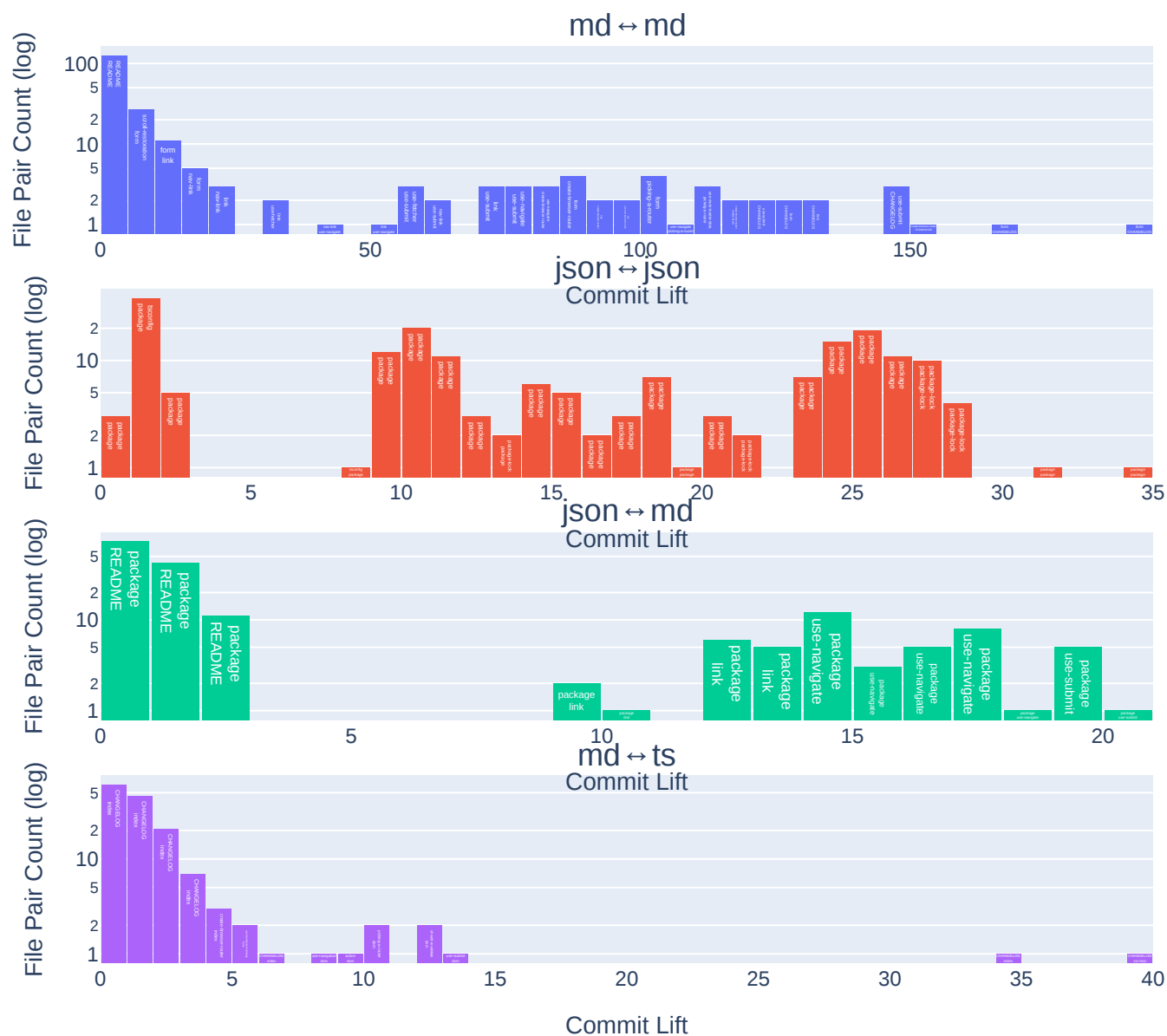
| fileExtensionPair | updateCommitMinConfidence | GroupRank | filePair | filePairWithRelativePath                                                                              |
|-------------------|---------------------------|-----------|----------|-------------------------------------------------------------------------------------------------------|
| 35                | md↔ts                     | 0.409091  | 6        | api-development-strategy↔router<br>docs/guides/api-development-strategy.md↔packages/router/router.ts  |
| 36                | md↔ts                     | 0.388889  | 7        | create-browser-router↔lazy-discovery-test<br>router.md↔packages/router/_tests_/lazy-discovery-test.ts |
| 37                | md↔ts                     | 0.384615  | 8        | README↔router<br>packages/router/README.md↔packages/router/router.ts                                  |
| 38                | md↔ts                     | 0.375000  | 9        | use-fetcher↔router<br>docs/hooks/use-fetcher.md↔packages/router/router.ts                             |
| 39                | md↔ts                     | 0.370370  | 10       | should-revalidate↔router<br>docs/route/should-revalidate.md↔packages/router/router.ts                 |





|    | fileExtensionPair | updateCommitLift | GroupRank | filePair                                  | filePairWithRelativePath                                                                 |
|----|-------------------|------------------|-----------|-------------------------------------------|------------------------------------------------------------------------------------------|
| 0  | md↔md             | 194.250000       | 1         | README↔README                             | examples/lazy-loading/README.md↔examples/route-objects/README.md                         |
| 1  | md↔md             | 166.500000       | 2         | README↔README                             | examples/modal/README.md↔examples/route-objects/README.md                                |
| 2  | md↔md             | 153.481481       | 3         | README↔README                             | examples/custom-filter-link/README.md↔examples/custom-link/README.md                     |
| 3  | md↔md             | 148.000000       | 4         | README↔README                             | examples/custom-filter-link/README.md↔examples/modal/README.md                           |
| 4  | md↔md             | 145.687500       | 5         | README↔README                             | examples/custom-query-parsing/README.md↔examples/route-objects/README.md                 |
| 5  | md↔md             | 133.200000       | 6         | README↔README                             | examples/auth/README.md↔examples/modal/README.md                                         |
| 6  | md↔md             | 129.500000       | 7         | README↔README                             | examples/custom-filter-link/README.md↔examples/route-objects/README.md                   |
| 7  | md↔md             | 124.320000       | 8         | README↔README                             | examples/auth/README.md↔examples/search-params/README.md                                 |
| 8  | md↔md             | 121.090909       | 9         | README↔README                             | examples/basic/README.md↔examples/modal/README.md                                        |
| 9  | md↔md             | 116.550000       | 10        | README↔README                             | examples/auth/README.md↔examples/route-objects/README.md                                 |
| 10 | json↔json         | 34.533333        | 1         | package↔package                           | examples/error-boundaries/package.json↔examples/notes/package.json                       |
| 11 | json↔json         | 31.393939        | 2         | package-lock↔package                      | examples/navigation-blocking/package-lock.json↔examples/navigation-blocking/package.json |
| 12 | json↔json         | 28.566176        | 3         | package↔package                           | examples/custom-filter-link/package.json↔examples/custom-link/package.json               |
| 13 | json↔json         | 28.406452        | 4         | package↔package                           | examples/lazy-loading/package.json↔examples/multi-app/package.json                       |
| 14 | json↔json         | 27.750000        | 5         | package↔package                           | examples/auth/package.json↔examples/custom-filter-link/package.json                      |
| 15 | json↔json         | 27.626667        | 6         | package↔package                           | examples/notes/package.json↔examples/scroll-restoration/package.json                     |
| 16 | json↔json         | 27.518750        | 7         | package↔package                           | examples/custom-filter-link/package.json↔examples/multi-app/package.json                 |
| 17 | json↔json         | 27.490114        | 8         | package↔package                           | examples/custom-query-parsing/package.json↔examples/lazy-loading/package.json            |
| 18 | json↔json         | 26.979167        | 9         | package↔package                           | examples/basic/package.json↔examples/custom-filter-link/package.json                     |
| 19 | json↔json         | 26.640000        | 10        | package↔package                           | examples/auth/package.json↔examples/search-params/package.json                           |
| 20 | json↔md           | 20.812500        | 1         | package↔README                            | examples/custom-filter-link/package.json↔examples/modal/README.md                        |
| 21 | json↔md           | 19.588235        | 2         | package↔README                            | examples/custom-link/package.json↔examples/modal/README.md                               |
| 22 | json↔md           | 19.028571        | 3         | package↔README                            | examples/modal/package.json↔examples/modal/README.md                                     |
| 23 | json↔md           | 18.500000        | 4         | package↔README                            | examples/basic/package.json↔examples/modal/README.md                                     |
| 24 | json↔md           | 17.760000        | 5         | package↔README                            | examples/ssr/package.json↔examples/auth/README.md                                        |
| 25 | json↔md           | 17.266667        | 6         | package↔README                            | examples/basic/package.json↔examples/auth/README.md                                      |
| 26 | json↔md           | 16.187500        | 7         | package↔README                            | examples/custom-filter-link/package.json↔examples/custom-filter-link/README.md           |
| 27 | json↔md           | 16.145455        | 8         | package↔README                            | examples/auth/package.json↔examples/basic/README.md                                      |
| 28 | json↔md           | 15.696970        | 9         | package↔README                            | examples/basic/package.json↔examples/basic/README.md                                     |
| 29 | json↔md           | 15.235294        | 10        | package↔README                            | examples/custom-link/package.json↔examples/custom-filter-link/README.md                  |
| 30 | md↔ts             | 39.176471        | 1         | create-static-handler↔ssr-test            | docs/routers/create-static-handler.md↔packages/router/__tests__/ssr-test.ts              |
| 31 | md↔ts             | 34.533333        | 2         | should-revalidate↔should-revalidate-test  | docs/route/should-revalidate.md↔packages/router/__tests__/should-revalidate-test.ts      |
| 32 | md↔ts             | 13.875000        | 3         | create-browser-router↔lazy-test           | docs/routers/create-browser-router.md↔packages/router/__tests__/lazy-test.ts             |
| 33 | md↔ts             | 12.465241        | 4         | api-development-strategy↔ssr-test         | docs/guides/api-development-strategy.md↔packages/router/__tests__/ssr-test.ts            |
| 34 | md↔ts             | 12.046512        | 5         | use-route-loader-data↔history             | docs/hooks/use-route-loader-data.md↔packages/router/history.ts                           |
| 35 | md↔ts             | 10.791667        | 6         | create-browser-router↔lazy-discovery-test | docs/routers/create-browser-router.md↔packages/router/__tests__/lazy-discovery-test.ts   |
| 36 | md↔ts             | 10.791667        | 7         | should-revalidate↔fetchers-test           | docs/route/should-revalidate.md↔packages/router/__tests__/fetchers-test.ts               |
| 37 | md↔ts             | 9.296825         | 8         | use-submit↔dom                            | docs/hooks/use-submit.md↔packages/react-router-dom/dom.ts                                |
| 38 | md↔ts             | 8.246462         | 9         | use-fetcher↔dom                           | docs/hooks/use-fetcher.md↔packages/react-router-dom/dom.ts                               |
| 39 | md↔ts             | 6.937500         | 10        | create-browser-router↔fetchers-test       | docs/routers/create-browser-router.md↔packages/router/__tests__/fetchers-test.ts         |

Co-Changed Files by their commit lift

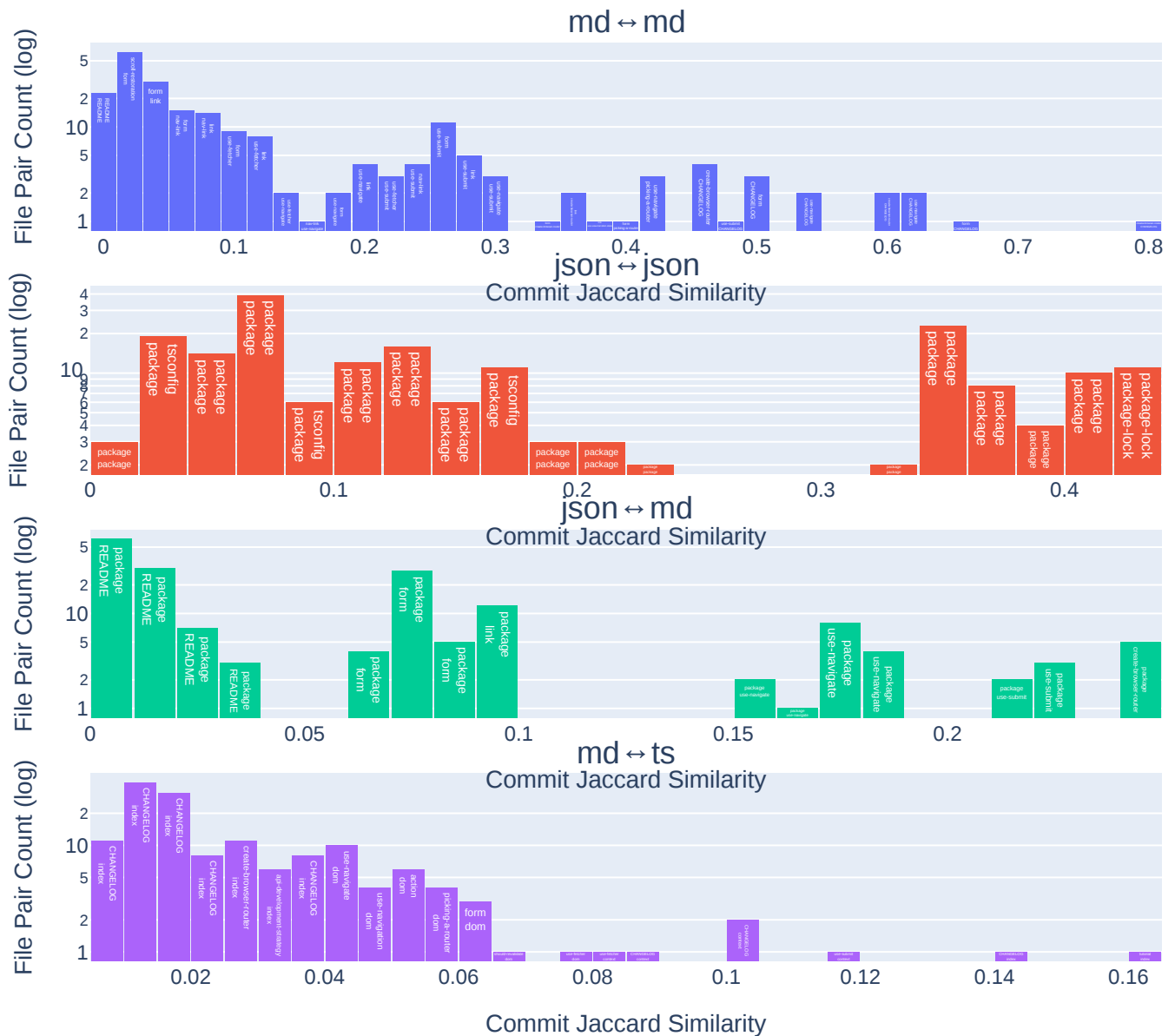


Files changed together by commit Jaccard similarity

|    | fileExtensionPair | updateCommitJaccardSimilarity | GroupRank | filePair                                  | filePairWithRelativePath                                                                          |
|----|-------------------|-------------------------------|-----------|-------------------------------------------|---------------------------------------------------------------------------------------------------|
| 0  | md↔md             | 0.800000                      | 1         | README↔README                             | examples/custom-filter-link/README.md↔examples/custom-link/README.mc                              |
| 1  | md↔md             | 0.666667                      | 2         | README↔README                             | examples/auth/README.md↔examples/search-params/README.mc                                          |
| 2  | md↔md             | 0.615385                      | 3         | README↔README                             | examples/auth/README.md↔examples/basic/README.mc                                                  |
| 3  | md↔md             | 0.600000                      | 4         | README↔README                             | examples/custom-filter-link/README.md↔examples/modal/README.mc                                    |
| 4  | md↔md             | 0.545455                      | 5         | README↔README                             | examples/auth/README.md↔examples/modal/README.mc                                                  |
| 5  | md↔md             | 0.500000                      | 6         | README↔README                             | examples/ssr/README.md↔examples/auth/README.mc                                                    |
| 6  | md↔md             | 0.470588                      | 7         | README↔README                             | examples/ssr/README.md↔examples/basic/README.mc                                                   |
| 7  | md↔md             | 0.461538                      | 8         | README↔README                             | examples/auth/README.md↔examples/custom-filter-link/README.mc                                     |
| 8  | md↔md             | 0.428571                      | 9         | README↔README                             | examples/basic/README.md↔examples/custom-filter-link/README.mc                                    |
| 9  | md↔md             | 0.400000                      | 10        | README↔README                             | examples/ssr/README.md↔examples/modal/README.mc                                                   |
| 10 | json↔json         | 0.434783                      | 1         | package↔package                           | examples/custom-filter-link/package.json↔examples/custom-link/package.jsor                        |
| 11 | json↔json         | 0.428571                      | 2         | package↔package                           | examples/auth/package.json↔examples/search-params/package.jsor                                    |
| 12 | json↔json         | 0.425532                      | 3         | package↔package                           | examples/auth/package.json↔examples/custom-filter-link/package.jsor                               |
| 13 | json↔json         | 0.420000                      | 4         | package↔package                           | examples/auth/package.json↔examples/basic/package.jsor                                            |
| 14 | json↔json         | 0.416667                      | 5         | package↔package                           | examples/basic/package.json↔examples/custom-filter-link/package.jsor                              |
| 15 | json↔json         | 0.408163                      | 6         | package↔package                           | examples/auth/package.json↔examples/custom-link/package.jsor                                      |
| 16 | json↔json         | 0.406780                      | 7         | package↔package                           | packages/react-router-dom/package.json↔packages/react-router/package.jsor                         |
| 17 | json↔json         | 0.400000                      | 8         | package↔package                           | examples/basic/package.json↔examples/custom-link/package.jsor                                     |
| 18 | json↔json         | 0.392157                      | 9         | package↔package                           | examples/basic/package.json↔examples/modal/package.jsor                                           |
| 19 | json↔json         | 0.386364                      | 10        | package↔package                           | examples/lazy-loading/package.json↔examples/multi-app/package.jsor                                |
| 20 | json↔md           | 0.245575                      | 1         | package↔CHANGELOG                         | packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.mc |
| 21 | json↔md           | 0.244009                      | 2         | package↔CHANGELOG                         | packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom/CHANGELOG.mc           |
| 22 | json↔md           | 0.242188                      | 3         | package↔CHANGELOG                         | packages/router/package.json↔packages/router/CHANGELOG.mc                                         |
| 23 | json↔md           | 0.241379                      | 4         | package↔CHANGELOG                         | packages/react-router-dom-v5-compat/package.json↔packages/react-router/CHANGELOG.mc               |
| 24 | json↔md           | 0.221957                      | 5         | package↔CHANGELOG                         | packages/router/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.mc                     |
| 25 | json↔md           | 0.220657                      | 6         | package↔CHANGELOG                         | packages/router/package.json↔packages/react-router-dom/CHANGELOG.mc                               |
| 26 | json↔md           | 0.218097                      | 7         | package↔CHANGELOG                         | packages/router/package.json↔packages/react-router/CHANGELOG.mc                                   |
| 27 | json↔md           | 0.216590                      | 8         | package↔CHANGELOG                         | packages/react-router-dom-v5-compat/package.json↔packages/router/CHANGELOG.mc                     |
| 28 | json↔md           | 0.189097                      | 9         | package↔CHANGELOG                         | packages/react-router-native/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.mc        |
| 29 | json↔md           | 0.188552                      | 10        | package↔CHANGELOG                         | packages/react-router-native/package.json↔packages/react-router-dom/CHANGELOG.mc                  |
| 30 | md↔ts             | 0.160494                      | 1         | use-submit↔dom                            | docs/hooks/use-submit.md↔packages/react-router-dom/dom.ts                                         |
| 31 | md↔ts             | 0.142857                      | 2         | create-static-handler↔ssr-test            | docs/routers/create-static-handler.md↔packages/router/__tests__/ssr-test.ts                       |
| 32 | md↔ts             | 0.118421                      | 3         | use-fetcher↔dom                           | docs/hooks/use-fetcher.md↔packages/react-router-dom/dom.ts                                        |
| 33 | md↔ts             | 0.104478                      | 4         | create-browser-router↔lazy-discovery-test | docs/routers/create-browser-router.md↔packages/router/__tests__/lazy-discovery-test.ts            |
| 34 | md↔ts             | 0.103448                      | 5         | should-revalidate↔should-revalidate-test  | docs/route/should-revalidate.md↔packages/router/__tests__/should-revalidate-test.ts               |
| 35 | md↔ts             | 0.088889                      | 6         | form↔dom                                  | docs/components/form.md↔packages/react-router-dom/dom.ts                                          |
| 36 | md↔ts             | 0.083333                      | 7         | api-development-strategy↔ssr-test         | docs/guides/api-development-strategy.md↔packages/router/__tests__/ssr-test.ts                     |

|    | fileExtensionPair | updateCommitJaccardSimilarity | GroupRank | filePair                        | filePairWithRelativePath                                                     |
|----|-------------------|-------------------------------|-----------|---------------------------------|------------------------------------------------------------------------------|
| 37 | md↔ts             | 0.075000                      | 8         | should-revalidate↔fetchers-test | docs/route/should-revalidate.md↔packages/router/___tests___/fetchers-test.ts |
| 38 | md↔ts             | 0.068376                      | 9         | create-browser-router↔index     | docs/routers/create-browser-router.md↔packages/router/index.ts               |
| 39 | md↔ts             | 0.061224                      | 10        | use-route-loader-data↔history   | docs/hooks/use-route-loader-data.md↔packages/router/history.ts               |

## Co-Changed Files by their commit jaccard similarity



## Find pairwise changed files with many highly ranked metrics

Find those pairwise changed files that have a high rank in many metrics by calculating a combined (weighted) score based on the ranks of each metric. This is useful to identify pairs of files that score high in most metrics, which indicates a strong co-change relationship.



|    | fileExtensionPair | filePair                                  | combinedMetricsScore | updateCommitCountExtensionRank | updateCommitMinConfidenceExtensionRa |
|----|-------------------|-------------------------------------------|----------------------|--------------------------------|--------------------------------------|
| 0  | json↔json         | package↔package                           | 13                   | 8                              |                                      |
| 1  | json↔json         | package↔package                           | 17                   | 8                              |                                      |
| 2  | json↔json         | package↔package                           | 17                   | 8                              |                                      |
| 3  | json↔json         | package↔package                           | 17                   | 8                              |                                      |
| 4  | json↔json         | package↔package                           | 17                   | 8                              |                                      |
| 5  | json↔json         | package↔package                           | 21                   | 7                              |                                      |
| 6  | json↔json         | package↔package                           | 21                   | 7                              |                                      |
| 7  | json↔json         | package↔package                           | 21                   | 7                              |                                      |
| 8  | json↔json         | package↔package                           | 23                   | 8                              |                                      |
| 9  | json↔json         | package↔package                           | 27                   | 7                              |                                      |
| 10 | json↔md           | package↔CHANGELOG                         | 33                   | 2                              |                                      |
| 11 | json↔md           | package↔CHANGELOG                         | 33                   | 2                              |                                      |
| 12 | json↔md           | package↔CHANGELOG                         | 35                   | 4                              |                                      |
| 13 | json↔md           | package↔CHANGELOG                         | 38                   | 4                              |                                      |
| 14 | json↔md           | package↔CHANGELOG                         | 38                   | 1                              |                                      |
| 15 | json↔md           | package↔CHANGELOG                         | 38                   | 4                              |                                      |
| 16 | json↔md           | package↔README                            | 38                   | 10                             |                                      |
| 17 | json↔md           | package↔README                            | 38                   | 10                             |                                      |
| 18 | json↔md           | package↔README                            | 38                   | 10                             |                                      |
| 19 | json↔md           | package↔README                            | 38                   | 11                             |                                      |
| 20 | md↔md             | README↔README                             | 15                   | 10                             |                                      |
| 21 | md↔md             | README↔README                             | 22                   | 12                             |                                      |
| 22 | md↔md             | README↔README                             | 22                   | 12                             |                                      |
| 23 | md↔md             | README↔README                             | 23                   | 10                             |                                      |
| 24 | md↔md             | README↔README                             | 25                   | 12                             |                                      |
| 25 | md↔md             | README↔README                             | 25                   | 12                             |                                      |
| 26 | md↔md             | README↔README                             | 27                   | 10                             |                                      |
| 27 | md↔md             | README↔README                             | 27                   | 10                             |                                      |
| 28 | md↔md             | README↔README                             | 29                   | 12                             |                                      |
| 29 | md↔md             | README↔README                             | 29                   | 15                             |                                      |
| 30 | md↔ts             | create-static-handler↔ssr-test            | 27                   | 19                             |                                      |
| 31 | md↔ts             | should-revalidate↔should-revalidate-test  | 28                   | 19                             |                                      |
| 32 | md↔ts             | create-browser-router↔lazy-discovery-test | 32                   | 15                             |                                      |
| 33 | md↔ts             | use-submit↔dom                            | 33                   | 10                             |                                      |
| 34 | md↔ts             | use-fetcher↔dom                           | 42                   | 13                             |                                      |
| 35 | md↔ts             | create-browser-router↔lazy-test           | 46                   | 19                             |                                      |
| 36 | md↔ts             | use-route-loader-data↔history             | 46                   | 19                             |                                      |

|    | fileExtensionPair | filePair                            | combinedMetricsScore | updateCommitCountExtensionRank | updateCommitMinConfidenceExtensionRa |
|----|-------------------|-------------------------------------|----------------------|--------------------------------|--------------------------------------|
| 37 | md↔ts             | create-browser-router↔fetchers-test | 60                   | 18                             |                                      |
| 38 | md↔ts             | api-development-strategy↔ssr-test   | 62                   | 19                             |                                      |
| 39 | md↔ts             | should-revalidate↔fetchers-test     | 62                   | 19                             |                                      |

## Pairwise changed files with pareto-optimal metrics

A pair (count, confidence, jaccard, lift) is Pareto-optimal if there is no other pair that is better or equal in all metrics and strictly better in at least one. In other words, it is not "dominated" by any other pair.

The frontier = the “best tradeoffs.”

## Pairwise changed files with pareto-optimal metrics - not considering file extensions

|    | filePair            | combinedMetricsScore | updateCommitCount | updateCommitMinConfidence | updateCommitJaccardSimilarity | update |
|----|---------------------|----------------------|-------------------|---------------------------|-------------------------------|--------|
| 0  | CHANGELOG↔CHANGELOG | 131                  | 117               | 0.422383                  | 0.267735                      |        |
| 1  | CHANGELOG↔CHANGELOG | 130                  | 123               | 0.431579                  | 0.272124                      |        |
| 2  | package↔package     | 74                   | 264               | 0.581498                  | 0.406780                      |        |
| 3  | CHANGELOG↔CHANGELOG | 128                  | 108               | 0.446281                  | 0.254717                      |        |
| 4  | package↔CHANGELOG   | 35                   | 93                | 0.395745                  | 0.242188                      |        |
| 5  | package↔package     | 13                   | 20                | 0.625000                  | 0.434783                      |        |
| 6  | package↔package     | 21                   | 21                | 0.600000                  | 0.428571                      |        |
| 7  | package↔package     | 21                   | 21                | 0.600000                  | 0.428571                      |        |
| 8  | package↔package     | 21                   | 21                | 0.600000                  | 0.428571                      |        |
| 9  | README↔README       | 15                   | 8                 | 0.888889                  | 0.800000                      |        |
| 10 | README↔README       | 29                   | 3                 | 0.750000                  | 0.428571                      |        |

## Pairwise changed files with pareto-optimal metrics - using ranks grouped by file extensions

|   | fileExtensionPair | filePair              | combinedMetricsScore | updateCommitCount | updateCommitMinConfidence | updateCommitJaccardSimila |
|---|-------------------|-----------------------|----------------------|-------------------|---------------------------|---------------------------|
| 0 | md↔ts             | CHANGELOG↔context     | 353                  | 3                 | 0.053571                  | 0.000                     |
| 1 | md↔md             | submit↔use-CHANGELOG  | 424                  | 3                 | 0.073171                  | 0.000                     |
| 2 | md↔ts             | CHANGELOG↔router-test | 332                  | 6                 | 0.033898                  | 0.010                     |

## WordCloud of git authors

|   | word              | frequency |
|---|-------------------|-----------|
| 0 | Michael Jackson   | 1832      |
| 1 | Ryan Florence     | 1119      |
| 2 | Matt Brophy       | 772       |
| 3 | Jimmy Jia         | 381       |
| 4 | Tim Dorr          | 326       |
| 5 | Chance Strickland | 180       |
| 6 | Mateusz Zatorski  | 124       |
| 7 | Logan McAnsh      | 99        |
| 8 | Dan Abramov       | 92        |
| 9 | Paul Sherman      | 77        |

Wordcloud of git authors

